

# SATQUERY AI — REMOTE SENSING MISSION REPORT

Indian Space Research Organisation (ISRO) | SIH 26167 Intelligence Dispatch

|                             |                                              |
|-----------------------------|----------------------------------------------|
| Natural Language Query:     | Quantify vegetation loss and built-up growth |
| Identified Workflow Intent: | CHANGE_DETECTION                             |
| Dispatched Specialist:      | ChangeDetectionEngine                        |
| Composite Confidence:       | 94.8% (HIGH)                                 |

## 1. Executive Analysis & Findings

**Assessment:** Bi-temporal comparative analysis reveals significant anthropogenic change: Built-up impervious surface expanded by +18.7% (approx. 6.97 sq km), primarily concentrated in eastern logistics and residential corridors. Vegetation canopy exhibited a concurrent decrease of -12.4% due to land clearing.

- Built-up area expansion: +18.7% over the observation interval
- Vegetation cover reduction: -12.4% (Agricultural/Woodland to Urban)
- Spatial co-registration validated with IoU >= 85%

## 2. 4-Signal Multi-Criteria Confidence Matrix

| Metric Signal | Component                            | Score (%) | Status   |
|---------------|--------------------------------------|-----------|----------|
| C_model       | Specialist Model Inference Certainty | 94%       | Optimal  |
| C_sensor      | Sensor Radiometric SNR & Quality     | 90%       | Optimal  |
| C_alignment   | Spatial Co-Registration (CRS & IoU)  | 100%      | Verified |
| C_resolution  | Ground Sample Distance Suitability   | 95%       | Optimal  |

## 3. Auditable Agent Execution Pipeline Trace

| Step | Action / Milestone                       | Tool Executed                | Latency | Status    |
|------|------------------------------------------|------------------------------|---------|-----------|
| 1    | Raster Ingestion & Metadata Parsing      | GeoTIFFProcessor             | 1.1 ms  | COMPLETED |
| 2    | Spatial Co-Registration & Alignment Vali | SpatialRegistrationValidator | 0.7 ms  | COMPLETED |
| 3    | Agentic Query Classification & Tool Rout | AgentClassifier              | 0 ms    | COMPLETED |
| 4    | Specialist Model Execution (ChangeDetect | ChangeDetectionEngine        | 0 ms    | COMPLETED |
| 5    | Evidence Fusion & 4-Signal Confidence Ra | CompositeConfidenceEngine    | 0 ms    | COMPLETED |